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$$f : \mathbb{R}^3 \rightarrow \mathbb{R} : (x, y, z) \rightarrow \frac{xy}{x^2 + y^2 + 1} - yz$$

$$Df = \begin{pmatrix} \frac{-x^2y + y^3 + y}{(x^2 + y^2 + 1)^2} & \frac{x^3 - xy^2 + x}{(x^2 + y^2 + 1)^2} - z & -y \end{pmatrix}$$

$$D^2f = \begin{pmatrix} \frac{2x^3y - 6xy^3 - 6xy}{(x^2 + y^2 + 1)^3} & \frac{-x^4 + 6x^2y^2 - y^4 + 1}{(x^2 + y^2 + 1)^3} & 0 \\ \frac{-x^4 + 6x^2y^2 - y^4 + 1}{(x^2 + y^2 + 1)^3} & \frac{2x^3y - 6xy^3 - 6xy}{(x^2 + y^2 + 1)^3} & -1 \\ 0 & -1 & 0 \end{pmatrix}$$

References